Nigusu Tigabu

Software Engineering Student | Backend Development & AI-Enabled Applications

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PROFESSIONAL SUMMARY

Software Engineering student (expected graduation 2027) with hands-on backend experience from a final-year team project: a Flask and PostgreSQL web system with JWT authentication and an integrated image-classification model for maize disease detection. Comfortable with REST APIs, Python, and basic React front ends, with exposure to explainable AI and big-data tools. Seeking internships, junior full-stack or Python roles, AI/ML-related positions, and freelance projects.

TECHNICAL SKILLS

Programming: Python (beginner), JavaScript (basic)

Backend: Flask, REST APIs, JWT authentication

Frontend & DB: HTML, CSS, React (basic), SQL, PostgreSQL

AI & ML: TensorFlow, MobileNetV2, Grad-CAM explainability

Big Data: Hadoop, Spark (academic project)

Tools: Git, GitHub, VS Code, Windows

PROJECTS

AI-Based Maize Disease Detection System

Backend Developer | Final-Year Team Project

- Developed backend services in Python and Flask to assist Ethiopian smallholder farmers in identifying maize leaf diseases.
- Integrated MobileNetV2 image classification with Grad-CAM visual explanations for model interpretability.
- Implemented PostgreSQL database and JWT authentication for secure farmer accounts and diagnosis logs.
- Contributed to localized disease advice, dual English/Amharic support, and PWA offline functionality.

Technologies: Python, Flask, PostgreSQL, JWT, TensorFlow, MobileNetV2, Grad-CAM, PWA

Water Quality Prediction

Academic Project

- Applied data analysis and machine learning models to analyze and predict water quality parameters.

Airline Delay Analysis

Academic Project

- Analyzed flight operational and delay data using Hadoop and Spark for a big-data course project.

BD Electronics & Computer Service Website

Personal Project

- Built a business website demonstrating responsive web design and modern frontend development.

Code: https://github.com/Mekdelawite/bd-electronics-website

Photo Studio Demo Website

Personal Project

- Designed a demo website for a photo studio emphasizing visual presentation and layout structure.

Code: https://github.com/Mekdelawite/photo-studio-demo

EDUCATION

Bachelor's Degree in Software Engineering

Expected Graduation: 2027

Mekdela Amba University, Ethiopia